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When Battery-Free Interfaces Support Design Changes Qualified Observations and Strong Static Controls

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Abstract

Battery-free design decisions depend on when energy is usable, which event state survives reset, and what reaches the receiver. Current success can omit the mechanism activated by a legal design change. We distinguish current-verdict from intervention sufficiency, give an elementary factorization condition, and connect storage and reset counterexamples to a source-bound observation interface. An earlier four-task comparison gives 8/12 service passes with average-energy inputs and 12/12 with each strong static or executable control; live observation completes seven chains and leaves five incomplete. We retain these failures and add deterministic cross-source unit, domain and type preflight before actual consumption. A new frozen comparison uses storage connection, reset placement and package-response interventions, each in two fresh contexts per arm. Complete information, conservative fixed rules and observation refinement each produce 6/6 correct predictions, decisions and selected-service passes. Typed observation consumption occurs in 1/6, 4/6 and 3/6 respectively; it is a path statistic, not the success criterion for direct baselines. Two fixed-rule episodes stop at the request cap after correct submissions. Refinement uses 50 requests versus complete information's 37, with equal source acquisition and a slightly lower fixed-rate fee estimate: there is no accuracy, acquisition or comprehensive cost advantage. Saved event studies additionally separate matched delivery from encoded-time accuracy. All results concern prescribed-source digital apparatus; physical qualification is unavailable and the original product remains unaccepted.

1 Introduction

A battery-free device must turn an intermittent supply into a particular service: capture an external event, preserve the relevant state, and deliver the required information at the required time. A source integral does not specify when a rail becomes ready, whether a transmit pulse causes a reset, or whether the pending event is still present when power returns. Intermittent execution mechanisms and type systems address parts of this consistency problem [1, 2]. The design consumer also needs an interface that exposes the laws and state transitions affected by its next legal change.

The information requirement is prospective. Two worlds can have identical current traces yet different unexcited storage clamps; reducing capacitance then separates their outcomes. Two event memories can likewise agree until a reset occurs. These witnesses explain why a current successful trace need not support intervention. They do not establish a universal advantage for live observations: a static description containing the relevant laws, parameters, initial state and intervention domain can already determine the answer. Our elementary distinguishability condition states that boundary without claiming a new general abstraction theory [3, 4, 5].

An observation interface provides an operational route from information to action. It acquires a source, selects or constructs a relation, checks its domain and units, and delivers a typed value to a design policy. Each step can fail independently. An equation can refer to the wrong source-local domain; a correct Boolean can fail to match numeric branches. Even a successfully consumed value can support a design already obtainable from complete static information. Admission, actual consumption, independent service and comparative benefit therefore require separate evidence [6, 7, 12].

The earlier matched digital study makes the strong static alternative explicit. On four fixed tasks with three repetitions per condition, rich-static, complete-input and executable-response controls each pass all twelve formal groups. Prospective live observation completes seven chains and leaves five incomplete. The equal-information JSON/table comparison also ties at twelve service passes and 65 requests per arm. We preserve these results and their original failed attempts rather than repair them into retrospective live successes [11, 12, 13].

The new study asks a narrower question after improving legal-input admission: with the same source access and execution permissions, does observation refinement improve judgments or costs on three frozen mechanism changes? Storage connection changes the usable electrical buffer; reset placement changes whether a volatile event survives until transmission; package response changes the time at which prescribed energy reaches the port. A deterministic preflight checks cross-source identity, unit conversion, applicability and consumer type before execution. It does not decide service. Each arm receives two fresh-context repeats per task, for eighteen formal attempts in total. The development legal-consumer example is researcher supplied and excluded from that denominator [23, 24, 25].

All three arms make six correct predictions and decisions and select six passing designs. Refinement consumes a typed observation in three attempts, uses 50 requests against complete information's 37, and acquires the same 89,218 source bytes. Its lower fixed-rate fee estimate does not establish comprehensive efficiency. Two fixed-rule episodes have correct direct submissions but retain request-cap stop status. This combination is informative: legal consumption can work without supplying an accuracy or acquisition advantage over a strong static control [24, 28].

The contribution combines the information condition, mechanism witnesses, executable source-to-consumer contract, and two finite comparisons that retain strong controls, failures and costs. The secondary event studies show why matched delivery and encoded-time accuracy must also remain separate. New component-level A/R results remain a finite negative result, and new event-time methods do not improve on the equal-history control. These boundaries prevent an interface-level result from becoming a claim of product acceptance, physical qualification or stable recursive improvement [18, 26, 27].

2 Related Work

Intermittent systems research addresses the consistency of execution across power failures. Alpaca organizes a program into tasks, privatizes shared data and commits task effects so interrupted execution can restart without exposing partial updates [1]. This establishes a concrete recovery mechanism; it does not make a current energy summary sufficient for a different capacitor or establish that an event reached its receiver. Our retained-state witnesses use an idealized latch and do not implement or evaluate Alpaca.

Curricle specifies intermittent idempotence requirements through a type system and information-flow qualifiers, checking programs against those requirements [2]. Its program-level consistency question complements our admission question: whether an observation has declared units and domain, is independently qualified on the supplied source, and reaches a compatible design consumer. Passing these interface checks does not prove intermittent idempotence, arbitrary program correctness or physical validity.

Counterexample-guided abstraction refinement (CEGAR) refines an abstraction when an abstract counterexample is spurious in the concrete system [3]. Our formulation shares the use of distinguishability to explain missing information. Its role is narrower: two legal digital worlds with the same representation require different verdicts after a specified intervention. Proposition 1 is an elementary factorization condition, not a new general theory of abstraction or a CEGAR algorithm. We supply neither a general model checker

nor an automated refinement guarantee. The empirical contribution is the source-bound comparison among complete static information, executable feedback and qualified live observations, including failed chains and receiver-side service conditions.

3 The information required by a design consumer

3.1 Current verdicts and legal interventions

Let T fix the external service and its evaluator, $w \in \mathcal{W}$ denote a world including device laws and state rules, and d_0 be a current design. The complete nominal verdict is $V_T(w, d) \in \{\text{PASS}, \text{FAIL}\}$. The representation $R_{d_0}(w)$ is the information legally available at the current design. It may contain parameters, records, source definitions or traces; it need not be a scalar. A legal intervention δ produces $\delta(d_0)$. Comparisons below use the same T and only interventions jointly legal in the compared worlds. A change in the success criterion is not a design intervention.

Current-verdict sufficiency requires

$$R_{d_0}(w) = R_{d_0}(w') \implies V_T(w, d_0) = V_T(w', d_0). \tag{1}$$

Intervention sufficiency on a declared common domain $\mathcal{D}$ requires the stronger implication

$$R_{d_0}(w) = R_{d_0}(w') \implies V_T(w, \delta(d_0)) = V_T(w', \delta(d_0)) \quad \text{for every } \delta \in \mathcal{D}. \tag{2}$$

Both are properties of a representation, consumer, world domain and task together. A field set sufficient for energy balance need not be sufficient for event retention or timestamp accuracy. Nor does either property promise that a finite-budget language-model procedure will use the information well.

Proposition 1 (Elementary factorization condition)**.** *On a fixed common legal domain, an exact prospective predictor $g(R_{d_0}(w), \delta) = V_T(w, \delta(d_0))$ exists if and only if the verdict for each δ is constant on every equivalence class of worlds sharing R_{d_0}.*

Proof. If g exists, identical representations and the same intervention give it identical arguments, so their required verdicts must agree. Conversely, if the verdict is constant on each class, define g by that class's verdict for the requested intervention. Constancy makes the definition independent of the representative world. This is an information condition; the proof does not establish an efficient implementation or calibrated world model. □

An interface can return UNKNOWN when its equivalence class contains both verdicts. Abstention then differs from an incorrect PASS or FAIL. We also distinguish this epistemic answer from the earlier study's INCOMPLETE, which denotes no admitted final execution. In the new batch, original runtime acceptance and actual submission are recorded separately: a budget-stopped episode can already contain a decision artifact (Section 9). Physical qualification UNKNOWN is a third boundary: it describes absent calibration evidence even when the digital verdict is well defined.

3.2 Refinement and the strong static alternative

Acquiring an observation O refines the representation to (R, O). It is informationally useful for a particular intervention only if it separates worlds whose prospective verdicts were previously ambiguous, or supplies another needed input to the consumer. Provenance, domain and unit checks establish admissibility of the value. They do not, by themselves, establish that the value distinguishes the relevant worlds or changes service.

A complete static model is an important limiting case. If its supplied parameters, laws and initial state determine the execution for every legal d, then executing or reasoning about that model defines an exact g on the stated ideal domain. A dynamic response can make a computation easier to inspect, and an observation can reduce the information a consumer must handle. Neither has an automatic informational advantage over

that complete static model. A finite-budget comparison must measure any computational or operational advantage rather than infer it from the word "executable."

This separates three interface families evaluated here. A current trace describes one realized behavior. A declarative static model additionally specifies how permitted changes affect that behavior. An executable response computes candidate-specific consequences. A qualified observation is a source-bound intermediate fact delivered through a typed edge. These are distinct interfaces, but their useful information can overlap completely. The static-grid checks and matched rich-static arm explicitly test that overlap [4, 12].

3.3 Service has several independent predicates

The BF service consumer combines energy feasibility, event capture, retained state, message validity and timing. An ordered retention witness requires the same event to be captured before a reset and first validly received afterward. A reset after delivery does not satisfy that witness. Similarly, matching an event identifier does not establish a correct encoded event time. For a confirmed event at t^*, receipt at t_{rx} and encoded stamp $\hat{t}$, an O-style service includes both

$$0 \le t_{\mathrm{rx}} - t^* \le L, \qquad |\hat{t} - t^*| \le \epsilon, \tag{3}$$

in addition to energy and state predicates. The later experiment uses $L = 10$ s and $\epsilon = 1$ s. The receive clock and the encoded stamp are different quantities. Passing the first inequality cannot replace checking the second [17, 18].

4 Mechanism counterexamples and their sufficient conditions

4.1 An unexcited storage clamp: E2

Two ideal worlds receive 20 μW for 2 s and face a 30 μJ service transaction at 2.5 s. Both start with zero stored energy, use $C = 10$ μF and must retain the energy associated with a 1 V minimum rail. Their clamps are 3 V and 5 V. The current maximum voltage is $\sqrt{8} = 2.828$ V, so neither clamp activates. Their complete current observed event and rail traces coincide and both services pass.

The common legal change sets $C = 4$ μF within the declared $[4, 12]$ μF domain. At 3 V, only 18 μJ can be stored; 22 μJ spills and only 16 μJ is available above the 1 V reserve. Service fails with a -14 μJ margin. At 5 V, all 40 μJ is retained and the margin is 8 μJ, so service passes. The task, source, requested transaction and common change are identical. Thus the current trace does not distinguish a mechanism needed for the prospective verdict [4].

An enriched static description resolves the ambiguity. Under the exact assumptions that all input precedes service, the capacitor is ideal and there is no leakage or intervening workload, define

$$M(C) = \min\left(E_0 + P_{\mathrm{in}} t_{\mathrm{in}}, \frac{C V_{\max}^2}{2}\right) - \frac{C V_{\min}^2}{2} - E_{\mathrm{service}}. \tag{4}$$

Service passes exactly when $M(C) \ge 0$. The first term is the stored energy after the source ends; the next is the required residual energy. With no other transaction between those times, their difference is exactly the budget available to service. The sufficient object is the functional $M(\cdot)$ and its parameters on the legal domain, not merely its value at the current C. Both worlds have the same current margin. A current scalar margin alone therefore retains the counterexample.

4.2 An unexcited reset transition: E3

The second pair captures event 1 at 1 s and attempts its single transmission at 4 s, before a 5 s deadline. A volatile pending memory and a retained pending memory both deliver in the current uninterrupted execution. The same legal change inserts a power outage from 2 to 3 s. Volatile memory clears the pending event and delivers none; retained memory delivers one. External event timing and the transmission schedule remain fixed. This witness changes a state transition rather than scaling an energy total [4].

The sufficient static state description must state capture, reset and send semantics and their order. If pending state survives each intervening reset, and the scheduled send can execute after restoration, the captured event remains available. If the only pending copy is cleared and no recapture is possible before sending, it does not. These are conditions of the ideal machine, not a claim that an arbitrary nonvolatile device has free writes, atomic persistence or a qualified energy budget.

Table 1: E2–E3 structural witnesses: both worlds pass before a common legal intervention, then separate. These are two exposed ideal pairs, not statistical repetitions.

Mechanism	World	Current	Changed	Changed events
Clamp	low-clamp	PASS	FAIL	0
Clamp	high-clamp	PASS	PASS	1
Reset memory	volatile	PASS	FAIL	0
Reset memory	retained	PASS	PASS	1

The saved analytic and numerical clamp implementations and the separate memory interpreter agree on eight current/intervention cases at both 1 ms and 0.5 ms settings: sixteen resolution-specific comparisons. Maximum reported energy discrepancy is 2.84×10^{-12} μJ. The enriched static model agrees on all 28 exposed grid points, and an additive static control agrees on five service costs spanning PASS and FAIL. Removing the unexcited clamp or reset-retention fact restores aliasing in these encodings. These checks establish software consistency under common ideal assumptions, not independent physical truth or universal necessity of a particular field name [4].

4.3 Connecting the witnesses to transient BF behavior

Seven separate saved BF controls exercise source timing, startup, loaded rail, reset and reception (Table 2). Their transient model uses ready/off thresholds, boot energy, finite TX current and duration, and the loaded-rail relation

$$V_{\text{load}} = \sqrt{2E/C} - I_{\text{TX}} R_{\text{ESR}}. \tag{5}$$

A failed energy or rail gate can reset the program and clear volatile pending state. The causal path is load to stored energy and loaded rail, then reset and program behavior. The source is prescribed, so no load-dependent source feedback is established [5].

Table 2: Seven saved digital BF reference controls. First reception after 100 ms fails the deadline; a passing rail gate without a received event is insufficient. Physical qualification is UNKNOWN throughout.

Control	Input (μJ)	RX (ms)	Service	Rail
Fast pulse	150	15.8	PASS	PASS
Slow pulse	150	54.8	PASS	PASS
High ESR	150	–	FAIL	FAIL
No energy	0	–	FAIL	PASS
Late pulse	150	105.8	FAIL	PASS
Latched reset	210	46.9	PASS	PASS
Volatile reset	210	–	FAIL	PASS

Fast, slow and late pulses each supply 150 μJ, yet their first receptions differ; the late one misses the 100 ms deadline. A 150 Ω ESR control prevents reception despite the same source integral. The no-energy control passes its rail check vacuously and fails service. The reset pair supplies 210 μJ with the same 40 μJ shunt and postponed transmission, but pending-event survival differs. Its retained mechanism is an explicitly ideal passive mechanical latch able to capture without stored electrical energy. It is not a model of a qualified commercial contact device or a powered NVM write.

All seven agree with their predeclared control expectations, including the intended failures. That agreement is different from seven service passes. The full input laws and source calibration limits accompany the saved

data; physical qualification remains UNKNOWN for every control. These transient examples connect the structural issue to BF operation without turning the two ideal witness models into a complete physical node.

5 Three frozen intervention mechanisms

The new comparison changes mechanisms in the existing transient B model, rather than selecting an easier service after seeing outcomes. All three families and the eighteen formal identities are fixed before their new runs. Each task has an immutable current input and one legal intervention. The service requires one matching eight-byte application frame by 100 ms, valid independent energy and receiver accounting, deduplication, and transient-voltage feasibility. The 150 ms trajectory uses a 0.1 ms step. The evaluator is the same within each task and across arms. Figure 1 reads the saved traces; no new simulation is used to construct it [24, 25].

5.1 Storage connection changes the buffer available to one transmission

The storage family changes two identical ideal cells from parallel to series while holding the PMIC clamp at 3.3 V. With per-cell capacitance $C_c = 11$ μF and ESR $R_c = 0.6$ Ω, the static equivalents are

$$(C_{\parallel}, R_{\parallel}) = (2C_c, R_c/2), \qquad (C_{\text{series}}, R_{\text{series}}) = (C_c/2, 2R_c). \tag{6}$$

Thus capacitance changes from 22 to 5.5 μF and ESR from 0.3 to 1.2 Ω. The prescribed 7.5 mW source lasts 20 ms in both inputs; the ideal passive event latch is unchanged. This mapping does not model cell balancing, voltage-rating changes, leakage mismatch or package parasitics.

Retain receives at 15.8 ms. Adopt boots earlier, begins transmitting at 4.3 ms, and resets at 6.1 ms before completing the two-millisecond frame. Four transient resets occur by 20.2 ms and no valid frame arrives. Independent energy conservation passes, while event reception and transient-voltage service fail. Earlier readiness and adequate source integral are therefore insufficient for this transaction. The stored energy above

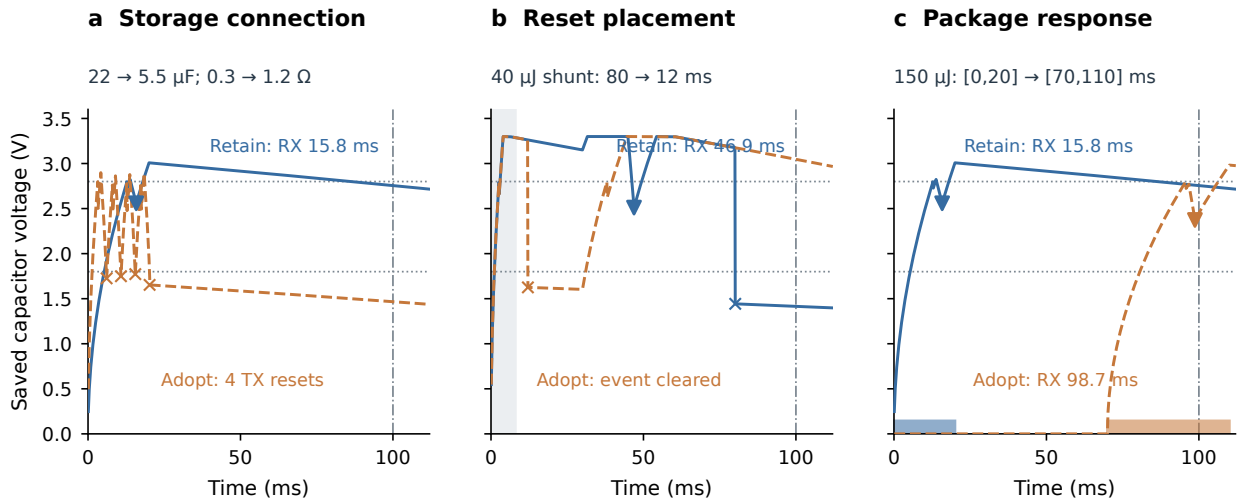


Figure 1: Frozen mechanism changes and saved post-step capacitor voltage. Blue solid is retain; orange dashed is adopt. Triangles mark actual first reception and crosses mark saved brownouts; dotted horizontal lines are the 2.8/1.8 V ready/off thresholds. The vertical dash-dot line is the 100 ms deadline. (a) Series-equivalent storage produces four interrupted transmissions. (b) Moving the shunt before TX clears volatile pending state; grey shading is the eight-millisecond contact window. The retain reset is after delivery. (c) Coloured strips mark prescribed source windows; the delayed intervention still receives at 98.7 ms. These are five distinct digital exact inputs, with the common retain trace reused in (a) and (c), not six physical experiments.

the off threshold, boot work, ongoing harvest and loaded-rail guard jointly govern completion. A proxy that compares only capacitor energy with TX work is not an exact failure theorem when the source can replenish energy during TX.

5.2 Reset placement changes event availability without changing its energy

The reset family retains 10 μF storage, a volatile event selector, a contact from 0 to 8 ms, and earliest TX at 45 ms. Two source pulses supply 210 μJ in total. The intervention moves the same 40 μJ shunt from 80 to 12 ms; it does not change its integral or insert another event. Both trajectories first boot at 2.7 ms and capture the event during the contact.

Retain completes delivery at 46.9 ms, before its 80.1 ms reset. Adopt resets at 12.1 ms, after contact has ended and before transmission is permitted. Its volatile pending state is cleared. A second boot at 38.1 ms cannot recapture an already ended contact; no frame is sent. Both independent energy and receiver-integrity checks pass, but the intervention's event-reception predicate fails. The discriminating object is the ordering of capture, state-clearing reset and send, together with the actual reset semantics. A post-delivery reset in retain does not demonstrate that volatile memory preserves an undelivered event across reset.

5.3 Package response changes usable time while preserving total input

The package family transforms the prescribed rectangular source from 7.5 mW on $[0, 20]$ ms to 3.75 mW on $[70, 110]$ ms. Both provide 150 μJ. Device parameters, event at zero, passive mechanical latch and deadline are unchanged. The late input first boots at 95.8 ms and receives at 98.7 ms, leaving 1.3 ms before the deadline; both retain and adopt pass.

This case is deliberately reported as a passing intervention. The shift does not imply failure merely because it resembles a late pulse. Conversely, total energy alone does not establish its tight reception margin. The passive latch allows the event to remain available before electrical startup. The input transformation is an assumed package-response law applied to the existing port model; it is not a measured mechanical transfer function or a new load-dependent source model. All family judgments remain digital and physically uncalibrated [24].

6 From a missing relation to an executed consumer

6.1 A bounded observation language and external qualification

The observation interface makes the information path explicit (Figure 2). The participant can list and acquire permitted sources, propose either a selected field or a constructed relation, and submit a typed design policy. A proposal states the source identifier, JSON Pointer paths, units, applicability domain and intended relation. Selecting a supplied field and constructing a new expression are recorded as different operations. Neither label alone establishes a new physical fact or a novel scientific relation [6, 9].

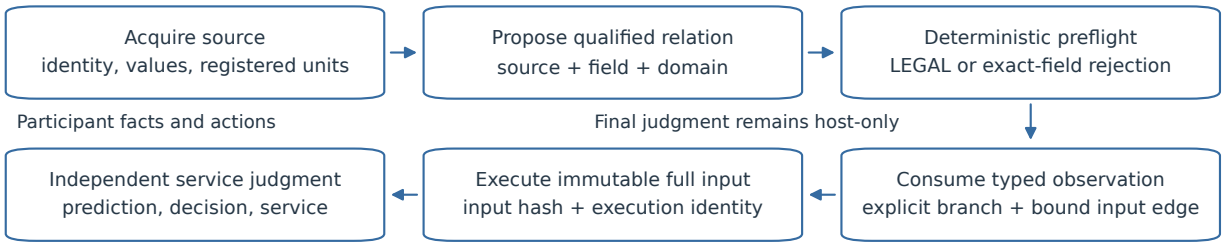


Figure 2: The observation-to-consumer chain. Rejection can stop the chain before any final service execution. Every attempted action and its cost remain recorded. A source-bound value or a consumed observation alone is not service or benefit evidence.

Construction uses a bounded JSON abstract syntax tree rather than arbitrary Python. The interpreter admits at most 64 nodes, depth eight, an 8 MB JSON source and 32,768 vector elements. A construction must contain a source field and an allowed operator. Unit dimensions must agree across operations; the declared output unit must match the root, and declared numeric domain constraints are checked against source values. Missing fields, unsupported domains and invalid arithmetic lead to rejection or UNKNOWN. These limits bound interpretation; they do not calibrate the supplied source.

Authority is outside the proposer. The host binds the original source bytes, proposal code and input, provider/run identity and actual accounting. An independent interpreter checks the source relation. A proposer cannot qualify its own output by supplying an authority record or a success label. The consumer receives only an admitted observation and the original policy, then chooses among at most three strictly typed equality branches. Boolean true and a numeric duration are different values. The complete selected input, observation edge and actual execution identity are preserved before the separate service verdict is read [6, 8, 9].

This design distinguishes acquisition from consumption. Echoing a value in a report is insufficient. A valid consumption receipt shows how that value entered the executed design. It still does not establish causal necessity: the same design might be chosen with complete static inputs, and an observation can be consumed without changing the design. A claimed benefit would require a separately frozen comparison of alternatives with the same task and allowance.

6.2 A concrete constructed relation and its actual consequence

An earlier actual E5 attempt acquired component, excitation and saved-activity records, then constructed an energy comparison. Its first proposal omitted the leading slash required by JSON Pointer. The original host counted that invalid proposal against the one-proposal limit; the model's already generated correction was then rejected at the cap. The live attempt remains rejected: no qualified observation, no real consumer execution, and no proposed-design scientific execution. Its six SETTLED requests and USD 0.0644115 upper-bound rate estimate remain charged [7].

A separately authorized record labeled `POSTHOC_HOST_ADAPTER_REEVALUATION` later evaluates that existing corrected call. It makes no new model request and does not replace the failed live attempt. The model's retained relation is

$$E_{\mathrm{TX}} = I_{\mathrm{TX}} \Delta t_{\mathrm{TX}} V_{\mathrm{out}} = 28.8\ \mu\mathrm{J}, \tag{7}$$

$$E_{\mathrm{reserve}} = \tfrac{1}{2} C (V_{\mathrm{ready}}^2 - V_{\mathrm{off}}^2) = 23.0\ \mu\mathrm{J}, \tag{8}$$

$$b = [E_{\mathrm{TX}} > E_{\mathrm{reserve}}] = \mathrm{true}. \tag{9}$$

The supplied record uses 8 mA, 2 ms, 1.8 V output, 10 μF capacitance and 2.8/1.8 V ready/off thresholds. The source-bound domain allows 4–22 μF and 0–45 ms earliest TX, with TX current fixed at the record's value. A separate Decimal interpreter agrees on the Boolean and verifies source dependence over legal capacitance values. The qualification criterion was revised post hoc from a paired-world-specific test to dependence across legal design values. That intervention and its scope remain explicit [8].

Before an admitted observation, the consumer keeps UNKNOWN and the original 10 μF, 45 ms design. Qualified Boolean true selects the model's original 15 μF, 45 ms branch. The actual full service changes from FAIL with no received frame to PASS with first reception at 46.9 ms. Both energy ledgers pass, so the energy flag alone cannot explain service. The selected design avoids a pre-deadline brownout; it does not demonstrate retention of a pending event across one. A complete static-information design already passes at 10 μF and 0 ms, receiving at 5.6 ms. The unused false branch's service is not established by these receipts [8].

The relation is therefore a concrete derived fact that reaches a real design consumer and has an executed engineering consequence. It is not prospective live-E5 success, a physical measurement, proof that capacitance must increase, or an advantage over complete static information. Its energy inequality omits startup work, losses and event order and cannot serve as a complete service evaluator. Later adapter-v2 regression records reject malformed syntax before charging the valid-proposal counter, while bounding syntax rejections and retaining a one-valid-proposal limit. Those new-version checks preserve the original failure [9].

7 Legal input before real consumption

7.1 Source identity, units and conditions are separate from a verdict

The additive `abc-e5-input-v1` protocol addresses the two admission failure classes exposed by the old E5 records. Each source leaf states its source identifier and JSON Pointer. The host binds source bytes, a fixed unit registry and the immutable current/intervention context. Each domain row names its own source, so a component capacitance and an excitation deadline need not incorrectly share a single source-local unit map. A source from another candidate context or an inapplicable source condition is rejected before execution [13, 23, 25].

Values are interpreted in the registered source units, then converted to compatible output and consumer units. For example, a farad-valued source can be bounded using microfarads without reinterpreting its stored number as a microfarad value. Numeric bounds and equality conditions are normalized before comparison. Incompatible dimensions, unavailable fields and invalid conditions produce a structured rejection naming the failing request field. The bounded expression interpreter and separate Decimal source-algebra check are reused; they do not read hidden service truth.

Boolean, numeric and string types remain distinct. A Boolean relation cannot implicitly select numeric-duration branches. If a designer intends that mapping, it must explicitly provide the true/false numeric values and units; prefix conversion then occurs before strictly typed branch matching. For example, an explicit true-to-2-ms mapping can match a 0.002 s branch. Omitting the mapping is an input-type error. The protocol repairs representational friction without silently changing a proposed consumer's meaning.

A LEGAL response means that the specified source relation, domain and consumer are consistent on that frozen input. It does not establish that the relation is necessary or sufficient for complete service over the whole declared range. Numerical agreement between two interpreters is not an independent proof of a scientific hypothesis. Final independent energy and receiver checks therefore remain outside preflight and outside the participant's final evaluator access.

7.2 The actual observation edge and the independent service edge

A legal relation reaches the consumer through a stored typed value and branch. The host records source bindings, normalized relation, selected decision, observation-to-input edge, full executed-input hash and execution identity. Only the selected immutable retain/adopt task can be dispatched. An input binding mismatch stops execution; a reported scalar without that edge is not counted as actual typed consumption. The same real executor and independent receiver/energy checks serve all arms. Fixture software tests are explicitly excluded from scientific evidence [25].

The separate development rehearsal demonstrates that this path is usable. A researcher supplies the earlier TX-versus-reserve relation, now bound to component and excitation sources with explicit conditions. Boolean true reaches the actual consumer and selects the frozen 15 μF intervention; reception occurs at 46.9 ms and selected service passes, while the current design fails. No model request is made. The submitted prediction is intentionally UNKNOWN, so prediction coverage is false even though typed consumption and selected service are established. The rehearsal is excluded from all eighteen formal attempts and establishes neither autonomous discovery nor comparative advantage [23].

Formal participants can also directly submit a decision. This route is legal for every arm, including refinement. A valid direct prediction and selected passing input are scientific outcomes even when no typed relation is consumed. Consequently the conjunction called `interface_decision_success` in the host record is a path-specific diagnostic. It is not the main success metric and cannot be used to penalize a complete-information baseline for choosing a correct design directly. Source legality, path use, scientific judgment, service and budget completion are reported independently in the new results.

8 Earlier matched digital experiment

8.1 Tasks, outcome and strong controls

The main comparison uses four fixed tasks. Pulse-fast is a static-sufficient control. Volatile-reset and latched-reset use the transient BF apparatus with different pending-state semantics. Their legal design variables are capacitance and earliest permitted transmission, under frozen task-specific ranges. Complete service uses independent energy and receiver checks plus the registered ordered-event analysis. PASS denotes admitted complete service, FAIL an admitted final design that fails it, and INCOMPLETE no admitted final execution [10, 11].

The fourth task is a preregistered C-lite single-query derivative. It supplies 20 μW ideal DC for 30 s, 1 μW idle load, zero initial energy and a 50 μJ reply. A query at 2 s must receive its matching reply by 7 s with maximum data age 5 s. Capacitance is 25–100 μF at an ideal 2 V ceiling; ESR, boot energy and boot delay are zero and harvest efficiency is one. Gates require query issuance, capture, matching token, causal order, deadline and freshness. The fixed token value 1 represents one virtual query, not a secure multi-query protocol or an RF environment. These are explicit synthetic assumptions [15].

Average-energy begins with the coarse source integral and average power. Rich-static and complete-input supply all input records and the full model definition. They have the same available facts in this registered comparison, making their tie an admissible control. The saved condition name for complete-input is `complete-raw`; it does not mean that the participant receives the full coupled physical trace. That trace is host-side audit material. Executable-response adds candidate-specific public energy ledgers, activity and packets. Live E5 begins coarse and actually acquires permitted records, proposes a field or relation, obtains external qualification and submits a typed consumer policy. No target expression, required-field list or answer-bearing relevance label is prefilled [10, 12].

E4 changes only the rich-static initial serialization, comparing JSON with a typed JSON-Pointer table. The table preserves nested containers, types and empty values. All twelve formal pairs have saved receipts of identical decoded initial packets and tools. Later source responses use common JSON in both arms. This is an initial-serialization comparison rather than a representation intervention across the whole interactive trajectory.

8.2 Repetitions, resource parity and cache exposure

Two pilot repetitions on volatile-reset precede the formal comparison. The registered one-time instability rule selects three formal repetitions, then freezes that count. Pilots remain separate development evidence. The full eight-condition matrix contains 112 unique groups and 140 actual episodes: 96 formal groups and 16 pilot groups. Parent method proposals account for additional episodes in two conditions. This paper displays six interface conditions, covering 72 formal and 12 pilot groups. Rich-static is also E4 JSON, and complete-input is also the fixed-method cell; these shared cells are counted once [10, 11].

Within each compared task, the scientific laws, service evaluator, legal design domain, reported model alias, initial cache and total resource caps remain fixed. Information access varies as declared by the condition. Each group has 24 provider requests, 40 tool calls, 1,200 s and at most two distinct design exposures, counting checks, response queries and confirmation. Acquisition, qualification and consumption do not add a free E5 budget. A parent proposal, when present, has an eight-request, 400 s sublimit within the group. The campaign caps are 3,000 requests, USD 200 in settled conservative estimates and 134,400 summed episode seconds. Failed groups are not automatically retried or replaced.

Every group starts from one immutable cache containing twelve exact-input responses and one registered derivative baseline. Private overlays isolate new job histories. A cache hit still consumes logical design exposure; it does not reveal another group's output before a query or enlarge the two-design cap. The records' term "physical cache" denotes the underlying digital execution cache and supplies no laboratory evidence. Fresh participant entries and the common cache identity are retained per group. The model is reported as `deepseek-flash`; this alias does not establish a pinned weight identity [11].

8.3 Preparation and negative controls

The structural preparation separately exports 64 inputs across four cases, four information levels, two formats and two observation regimes. Thirty-two format pairs preserve decoded facts, assets, tools and resource limits. These are preparation checks, distinct from actual model trials. Fixed parser checks abstain on all four current-only cases and correctly judge all four sufficient-information cases in both supplied serializations. They do not measure language-model comprehension [4, 6].

Negative controls retain the enriched static solution, complete raw model, uninformative observation, exhausted acquisition budget, illegal design changes and tampered source binding. A researcher-specified required-field selector and a researcher-supplied headroom expression are construction baselines, not autonomous discoveries. Preparation outcomes are not substituted for later live completion, and software validation of a consumer edge is not promoted to hardware acceptance.

9 New equal-access intervention comparison

9.1 Frozen tasks and strong baseline information

The new finite list comprises the three tasks in Section 5, three assigned arms and two fresh-context repeats per task: eighteen formal attempts, six per arm. It is not a new eighteen-task benchmark. The mechanism families, exact retain/adopt inputs, arm instructions and repeat identities are frozen before new task execution; sampling phase is reserved and excluded. Neither the researcher-provided development request nor another arm's answer is placed in formal context. No task is replaced after its outcome [24, 25].

Complete information receives the full source payload initially and can use all tools. The fixed-rule control checks ready/off and loaded-rail feasibility, usable energy during the operation, event availability across any reset, and deadline. It is a conservative classical rule set, not a deliberately weak average-energy control. Refinement is instructed to identify an activated mechanism, acquire the corresponding sources and conditions, and express a relation that drives the typed consumer. It can also submit directly. The assigned arm is retained regardless of the actual path taken.

All arms have the same complete source catalogue and execution permissions: component records, excitation records, full current/intervention inputs and the unchanged numerical model law. Every arm can acquire the entire catalogue in one call and can use preflight, checks or direct submission. Initial context serialization and the method instruction differ by design; identical source availability is not a claim of byte-identical initial prompts. The audit checks source payload parity and within-task scientific inputs. Initial payload volume and subsequent acquisition are therefore accounted separately.

9.2 Common resources, hidden checks and outcome definitions

The common ceilings are twelve model requests, forty tool calls, 900 s and two distinct immutable designs. Source acquisition, invalid attempts, preflight and rework all consume the same episode allowance. Maximum output is fixed at 65,536 tokens for this new batch after backend qualification; it does not rescue or remove the four older 32,768-token censored O episodes. All formal participants use fresh contexts and the reported alias `deepseek-flash`. The underlying weight identity remains UNKNOWN [25].

The participant predicts PASS, FAIL or UNKNOWN for the intervention and chooses adopt or retain. The registered decision rule is adopt if the intervention passes, otherwise retain; retain is therefore not automatically the correct answer for the passing package task. Independent host evaluation checks both fixed inputs after the public endpoint closes, filling an unexecuted input through the same exact-input executor. No hidden final verdict is a tool action.

We retain five distinct measurements: actual artifact submission; prediction coverage and correctness; decision correctness and selected service; actual typed consumption; and original normal-completion status. A request-cap stop can follow an actual submission, so an incomplete runtime status does not imply that no artifact exists. Budget-compliant service requires both the original normal-completion status and an independent service pass. Every assigned row, including stopped rows, remains in the six-per-arm denominator.

9.3 Execution and cost accounting

The exact-input cache preserves one canonical raw artifact per input. Each episode's host evaluation exposes two inputs, regardless of whether one or both are cache hits. We report evaluated-input references, cache reuses and new scientific calculations separately. Later arms inherit a warmer shared cache; zero new calculation in those arms is a schedule effect, not an algorithmic advantage. A cache hit supplies no free extra candidate or automatic retry.

Cost views retain requests, input/output/reasoning tokens, initial payload volume, acquired bytes, preflight/reject counts, tool time and total wall time. Host evaluation is included in total episode wall time but lacks a separately recorded duration. We do not infer an exact host-only time by subtraction. The reported fee is a conservative fixed-rate estimate on settled usage, not an invoice. Codex preparation cost remains UNKNOWN. With only three fixed tasks and two contexts per task, comparisons are descriptive; no significance, population equivalence or generalization claim is made.

10 Earlier results: the strong static control

10.1 The complete-service comparison

Table 3 reports every formal group. Average-energy yields 8 PASS and 4 FAIL. Rich-static, complete-input, executable-response and E4 table each yield 12 PASS. Live E5 yields 7 PASS and 5 INCOMPLETE; it has no admitted final service FAIL. Its completion rate is 7/12, not a conditional 7/7. Missing final executions and submitted service failures remain different outcomes with the same full denominator.

Table 3: Formal matched interface comparison: four fixed tasks, three repetitions per task, twelve groups per arm. P/F/I are PASS, submitted FAIL and INCOMPLETE. Every incomplete group and its requests remain included.

Condition	P	F	I	Requests	Requests/group
Average energy	8	4	0	75	6.2500
Rich static / E4 JSON	12	0	0	65	5.4167
Complete input	12	0	0	64	5.3333
Executable response	12	0	0	66	5.5000
Live observation (E5)	7	0	5	80	6.6667
E4 typed table	12	0	0	65	5.4167

Table 4: Formal P/F/I out of three per task. Each strong arm separately denotes rich static, complete input, executable response and E4 table; none has an omitted failure.

Fixed task	Average energy	Live E5	Each strong arm
Pulse-fast	3/0/0	2/0/1	3/0/0
Volatile reset	1/2/0	2/0/1	3/0/0
Latched reset	1/2/0	1/0/2	3/0/0
C-lite query derivative	3/0/0	2/0/1	3/0/0

The coarse arm's observed deficit occurs in the two reset tasks, each with one PASS and two FAIL. Pulse-fast and the C-lite derivative each have three PASS. Every strong arm passes all three repetitions on each task (Table 4). This finite result is consistent with the value of supplied timing and state information beyond the coarse energy summary. It establishes no service advantage for executable-response over rich-static information. The saturated controls also do not prove equivalence on harder or previously unused tasks.

The two pilot repetitions are not pooled with the formal twelve. On their single volatile-reset input, average-energy has one PASS and one FAIL; each other displayed arm has two PASS. E5's two pilot completions include one selection and one construction. These are the development observations preceding the one-time formal-repeat freeze, not additional formal successes or independent tasks [12, 10].

10.2 What the seven prospective E5 chains did

Every formal E5 completion contains an actual prospective proposal, external source qualification and a typed observation consumed by the design policy. Six select existing fields and one constructs a relation. Five use component records and two excitation records. Six change the initial design and one does not. All seven final designs pass independent service, and all seven deliver before any reset. Their successes therefore establish neither pending-state survival across a reset nor the necessity of the acquired observation. New physical measurements number zero [12, 11].

Table 5: Disposition of all twelve formal E5 groups. Repeated rejected actions are not additional groups; absent executions are retained as INCOMPLETE.

Disposition	Groups	Final service
Qualified selection, consumed	6	PASS
Qualified construction, consumed	1	PASS
Source-unit binding rejected	4	INCOMPLETE
No typed consumer match	1	INCOMPLETE

Four incomplete groups retain `SOURCE_UNIT_UNAVAILABLE` in their original gateway records. The code identifies a submitted source/field/domain unit-binding problem; it does not imply that a physical quantity does not exist. In the pulse-fast example, the selected packet field `/packets/0/application_bytes` is unit-mapped in the acquired saved-activity source, while the declared domain also references deadline, TX-duration and readiness fields absent from that source's unit map. A latched-reset proposal selects a packet field from the model-law source, whose mapped fields describe model identity and Python definition. The original generic error does not name the first rejected term. We diagnose the source-local mismatch from saved proposals and logs; no repaired proposal is evaluated [13].

The fifth incomplete group records `NO_TYPED_CONSUMER_MATCH` on the C-lite derivative. Its equality expression outputs a Boolean while the consumer branches match numeric durations 0.002 and 0.001. A value can therefore be syntactically meaningful yet fail to enter the intended design edge. Other syntax and valid-proposal-cap errors remain charged actions inside these groups, not extra independent trials or free rescue attempts. Table 5 counts groups once.

The rejected original E5 attempt and its separately authorized post-hoc adapter chain remain outside these twelve formal groups. The post-hoc record demonstrates engineering linkage of an already generated relation. The seven later formal completions demonstrate prospective operation under a different registered protocol. Relabeling the former as a successful live discovery would erase both its original admission failure and the host intervention [7, 8].

10.3 Equal information and acquisition overhead

E4 gives identical complete-service and request totals: JSON and the typed table each pass 12/12 using 65 provider requests, or 5.4167 per group. The retained USD and wall receipts differ (Table 6), but this small comparison isolates neither all latency determinants nor the representation of later acquisitions. It supplies no observed service or request advantage and no causal format-speed or format-cost conclusion [12].

Table 6: E4 equal-information initial serialization. Later acquisitions use common JSON. Cost and summed group wall time are descriptive; neither establishes a format advantage.

Initial packet	PASS	Requests	USD	Wall (s)
JSON	12/12	65	0.4267200	676.5
Typed table	12/12	65	0.5145681	611.4

The earlier finite observation selector chooses the smallest eligible source using researcher-supplied required-field metadata, without service outcomes. It reads the same 103 bytes and facts as the fixed mechanism-spec

Table 7: Finite observation selection on four exposed cases. Read and audit bytes are Python payloads, not disk I/O. Host timings are single historical measurements, not latency benchmarks; selected and mechanism-spec contain the same facts.

Observation	Correct	Read B	Audit B	Wall ms	Model runs
uninformative	0/4	4648	4648	0.289	0
mechanism-spec	4/4	103	103	0.260	0
fixed-rich	4/4	4747	4747	0.282	0
full-raw	4/4	8544	8544	0.333	0
executable-dynamic	4/4	8544	8544	1.886	4
selected	4/4	103	103	0.236	0

baseline; both correctly judge four cases. Rich-static, complete raw and an executable response also correctly judge all four. The historical timings in Table 7 include acquisition and source audit, plus selection when used. They are single-host payload measurements with uncontrolled cache state, not storage-I/O or monetary measurements. Thus lower payload volume relative to a full model is not an independent selection-algorithm gain [4].

A deterministic compiler also constructs the researcher-specified headroom observation. Its saved outputs of -14 and 8 μJ agree with twenty exposed clamp-grid checks and two fresh-process executions. Construction took 0.239 ms, reference qualification 13.017 ms and the two fresh processes 37.730 ms combined in those retained receipts. Because the equation and relevant fields were supplied by a researcher, this is executable construction machinery, not an autonomous discovery.

10.4 Service success versus an ordered retention witness

Four formal participant groups select a pending-event-across-reset result: two complete-input groups, one proposed-method group and one E4 table group. They make four fresh choices but reuse one distinct historical digital execution key. All select 10 μF and earliest TX at 45 ms. The same event is captured from 3.7 to 7.9 ms, a brownout occurs at 12.1 ms, and first valid reception follows at 46.9 ms. The fixed ideal passive latch provides the retention mechanism [14].

All twelve rich-static, all twelve executable-response and all seven successful E5 groups instead deliver before any reset. Their complete service passes are valid for their tasks but are not evidence for the stronger ordered-retention property. Conversely, four selections of one response are not four independent physical experiments or four newly invented retention mechanisms. The ordered audit creates zero new scientific executions and retains physical qualification UNKNOWN.

11 New results: legal consumption without an outcome advantage

11.1 All three assigned arms reach the correct scientific decision

Table 8 uses all eighteen frozen attempts. Each arm has six actual decision artifacts, six correct predictions, six correct adopt/retain decisions and six independent selected-service passes. All intervention predictions have coverage. There are no wrong adoptions or rejections. Every context rejects the storage and reset interventions and adopts the passing package intervention. The complete-information baseline therefore leaves no observed scientific-accuracy deficit for refinement to remove on this finite task list [24, 28].

Normal budget completion is a separate outcome. Complete information and refinement each finish normally in six attempts; fixed rules finish normally in four. The first fixed-rule repeats on reset placement and package response each reach twelve settled requests and retain `INCOMPLETE_OR_STOPPED`. Before stopping, both have an actual direct submission with a correct prediction and decision and selected-service PASS. They are neither missing submissions nor normally completed episodes. Their full costs and the original stop labels remain in the table denominator.

Table 8: New formal outcomes, each out of six assigned attempts (three tasks, two fresh contexts each). Actual submission is distinct from normal budget completion. Pred., Dec. and Svc. denote correct intervention prediction, correct adopt/retain decision and selected-service PASS. Typed is only a path statistic.

Assigned arm	Actual sub.	Pred.	Dec.	Svc.	Budget	Typed
Complete information	6/6	6/6	6/6	6/6	6/6	1/6
Fixed conservative rules	6/6	6/6	6/6	6/6	4/6	4/6
Observation refinement	6/6	6/6	6/6	6/6	6/6	3/6

Actual typed consumption occurs in one complete-information attempt, four fixed-rule attempts and three refinement attempts. The other decisions are legal direct submissions. In particular, three refinement attempts do not complete the proposed typed path but do reach the correct engineering decision. They remain in the assigned refinement arm; no replacement episode is run. The 1/4/3 counts describe use of an optional path, not relative scientific success. Applying the path-specific conjunction to all arms would incorrectly penalize a strong direct baseline.

11.2 What a formal typed observation establishes

The refinement reset example binds event/contact and transmission timing from complete inputs to the shunt time in the excitation source. Its Boolean result is false and is actually read by the retain branch. The stored edge identifies both source hashes, the typed value and the executed current-input hash. The independent saved trajectories then establish why retain passes and adopt fails: the 12.1 ms intervention reset clears the volatile event after the contact ends, whereas the current input sends before its late reset.

The passing package example also consumes a Boolean, now true, and selects adopt. Its proposed relation combines pulse start, deadline, boot delay, TX duration, power and efficiency with the energy needed to reach the ready threshold. The independent executor supplies the actual 98.7 ms reception. The source calculation is a qualified intermediate fact; the final service check additionally covers event availability, loads and reception. Correct service on this one input does not prove that the proxy is sufficient over every numeric bound in its declared domain.

The storage example consumes false and retains parallel storage. Its energy-window proxy correctly accompanies that decision on the frozen input, but does not replace the full transient model. In particular, harvest during TX, boot-step overshoot and the ESR guard must be considered before promoting such a proxy to an exact criterion. The raw proposal and its explanatory text are retained as participant outputs; interpreter agreement and a correct selected outcome do not validate every explanatory statement. No formal attempt supplies a new physical measurement or establishes universal autonomous discovery [24, 25].

11.3 More requests, equal acquisition and a mixed cost result

Table 9 shows 37, 59 and 50 requests for complete information, fixed rules and refinement. Each acquires exactly 89,218 logical source bytes over six attempts. Refinement therefore has no observed acquisition-volume gain and uses thirteen more requests than complete information. Preflight calls number 15/34/20 and rejected actions 9/27/14. Improving the admission schema does not remove operational errors or make extra interaction free.

The corresponding conservative fee estimates are USD 0.5999520, 0.8165511 and 0.5593839. The calculation uses input and output usage at fixed rates:

$$\widehat{C}_{\mathrm{USD}} = 0.30\frac{N_{\mathrm{input}}}{10^6} + 1.20\frac{N_{\mathrm{output}}}{10^6}. \tag{10}$$

Reasoning tokens are an output-accounting field, not an additional additive budget. Refinement uses fewer total input and output tokens than complete information in these receipts despite its larger request count. That yields the slightly lower fee estimate; it does not make every cost dimension better. Rates, response

Table 9: New formal resource totals over six attempts per arm. Tokens are total input/output usage; reasoning tokens are not added a second time. USD is the conservative fixed-rate estimate, not an invoice. Acquired bytes are logical source-response bytes, not disk I/O or total context volume.

Assigned arm	Requests	Input tokens	Output tokens	Est. USD	Acquired B
Complete information	37	1,207,152	198,172	0.5999520	89,218
Fixed conservative rules	59	1,816,101	226,434	0.8165511	89,218
Observation refinement	50	1,157,649	176,741	0.5593839	89,218

length and initial payload volume all enter this descriptive comparison. The estimates are not invoices and exclude unmetered preparation.

All arms have twelve evaluated-input references. Complete information accounts for four new scientific computations and eight cache reuses; fixed rules and refinement each reuse twelve cached inputs and add zero computations. Five distinct exact inputs underlie the three mechanisms because two tasks share the same current design. The cache is warmer for later arms, so the zero is not an algorithmic compute saving. Tool time is separately reported from total wall time, and the unrecorded host-only duration remains unavailable.

The B formal batch totals 146 settled requests, USD 1.9758870 in these estimates and 2552.7 summed episode seconds. The complete finite cycle has 48 model episodes and 307 settled requests with an estimate of USD 5.5299957; this already includes B, A and C formation/validation/formal work and must not be added to their subtotals. The developer rehearsal has zero model requests and remains outside eighteen [24, 25].

11.4 Relation to the earlier negative comparison

The new protocol establishes actual legal consumption under frozen new mechanisms and preserves all original source/type failures. It does not retroactively change the earlier seven-pass/five-incomplete E5 result, and the new six-per-arm scientific outcome cannot be read as a controlled before/after effect of preflight: task list, resource limits and protocol version differ. Both studies support the same bounded comparison. Complete

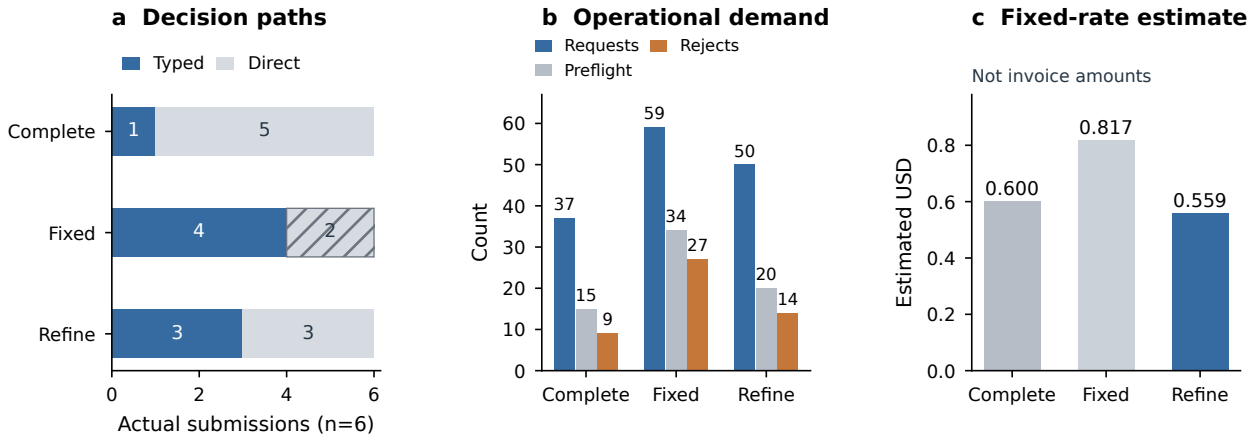


Figure 3: New formal path and cost comparison, all six assigned attempts per arm. (a) Typed and direct submissions both count as actual artifacts; hatching marks the two fixed-rule budget stops after direct submission. (b) Requests, preflights and rejects are distinct operational counts. (c) Conservative fixed-rate fees, rounded here to three decimals. All arms independently have six correct predictions, six correct decisions and six selected-service passes. Bars are finite totals across three fixed tasks and two fresh contexts per task; no error bars, significance or population estimate is implied.

Table 10: Operational and execution accounting. Initial B is summed initial payload bytes; PF/rej. count preflights/rejected actions. Wall includes host evaluation; no separate host-only duration was recorded. Eval/cache/new count evaluated-input references, cache hits and new scientific computations, respectively. Cache order prevents an algorithmic-savings interpretation.

Arm	Initial B	PF	Rej.	Tool s	Wall s	Eval	Cache	New
Complete	175,414	15	9	0.376	870.3	12	8	4
Fixed	62,054	34	27	0.162	926.7	12	12	0
Refine	61,850	20	14	0.176	755.7	12	12	0

Table 11: Every new formal attempt, with no deletion or replacement. C/F/R denote complete information, fixed rules and refinement. S is normal ACTUAL_SUBMITTED completion; X is original INCOMPLETE_OR_STOPPED. Every row has an actual artifact, correct prediction/decision and selected-service PASS; T indicates actual typed use.

Mechanism	Arm	Repeat	Status	T	Requests	PF	Rejects
package response	C	1	S	0	7	3	2
package response	C	2	S	0	7	4	3
package response	F	1	X	0	12	9	7
package response	F	2	S	1	11	6	5
package response	R	1	S	1	6	3	2
package response	R	2	S	0	10	5	4
reset placement	C	1	S	0	4	2	1
reset placement	C	2	S	0	6	2	1
reset placement	F	1	X	0	12	8	7
reset placement	F	2	S	1	6	2	1
reset placement	R	1	S	0	10	4	2
reset placement	R	2	S	1	8	2	1
storage connection	C	1	S	1	7	2	1
storage connection	C	2	S	0	6	2	1
storage connection	F	1	S	1	10	6	5
storage connection	F	2	S	1	8	3	2
storage connection	R	1	S	1	7	3	2
storage connection	R	2	S	0	9	3	3

static information is a successful control, and the live refinement route has not demonstrated additional scientific success. The new study further shows equal source acquisition, more requests than the strongest control, and a mixed rather than comprehensive cost advantage.

12 Event-time service exposes another interface boundary

12.1 The original O service and its strong fixed control

The O study extends the digital apparatus to vibration start, active keepalive and stop service. O0 is a 21,600 s task with zero initial capacitor energy, $C = 15$ mF and a prescribed source associated with a 15 cm^2 photovoltaic area. Its six intervals have nominal light levels 200/0/50/200/0/50 lux at boundaries 0/120/150/180/300/330/360 min. The corresponding electrical inputs are separately prescribed as 90/0/20/90/0/20 μW; the 3500 K label is metadata and lux is not converted into watts. RMS vibration is 0.08 g in the four declared activity intervals and 0.005 g otherwise. These inputs are uncalibrated digital assumptions [17].

Start requires RMS continuously above 0.05 g for 3 s; stop requires RMS continuously below 0.01 g for 60 s. Equality resets confirmation. The confirmation instant is the event time. Each start and stop must arrive within 10 s and carry an encoded timestamp within 1 s; active keepalive gaps must not exceed 120 s and idle

false events fail. The first dark interval contains one required active interval, so darkness cannot simply be excluded from service.

The inherited ideal energy path has 0.8 harvest efficiency, 0.38 V/ 3 μW cold-start gates, 2.7/2.3 V ready/off thresholds, 3.3 V maximum capacitor voltage, 5.5 V rating and 0.3 Ω ESR. It supplies 1.8 V with 0.3 V dropout, 1/5 μW off/sleep power, 100 μJ boot and 130 μJ atomic TX transactions. Added RMS costs are 0.5 μW standby, 1 μJ wake, 20 μJ per successful sample and 5 μJ per committed write. The write is not charged at every poll. These proposed costs do not qualify a vibration sensor, PMIC, clock, NVM or RF receiver.

O0 compares a free legal candidate, a fixed 60 s periodic snapshot control and a fixed rich-state start/stop machine. They share the same task, source, hardware and event evaluator. The candidate and rich-state control submit the same one-second, NVM, `confirmed_events` program with the same thresholds, one-second retry and zero backoff. They share an exact execution key and response, match eight events, deliver 18 accepted packets and pass the nominal O predicate. The periodic control delivers 348 accepted packets and passes energy, but matches no required event. It fails service [16].

Table 12: Original O0 development task. Candidate and rich-state control share one exact design and response. Self-judgment is scoped separately from the host O event predicate.

Arm	Poll (s)	Matched/8	Host O	Self-judgment	Requests
candidate	1	8/8	PASS	UNKNOWN	8
fixed-periodic	60	0/8	FAIL	FAIL	5
rich-state	1	8/8	PASS	PASS	5

The candidate checks one-second sampling, then 1.5-second sampling, and ultimately submits the former. In the retained 1.5-second alternative, the first start is observed at 4204.070423 s and stamped 4204.139063 s against truth 4203 s. The stamp error 1.139063 s exceeds tolerance even though energy and delivery can appear adequate. This checked alternative is preserved in the same episode's exposure and cost history, not treated as a later repair to its accepted design [16].

The candidate's own judgment is UNKNOWN. Its original explanation concerns unobserved dark intervals after brownout, absent gap-report messages, inherited A/R first-report and dark-report metadata, and unavailable source/hardware calibration. In particular, inherited metadata calls for reception by 1800 s and dark reporting every 600 s, while the first confirmed O event is later. The host's O predicate asks a narrower event-service question and returns PASS. Its saved disagreement flag is nominal to that predicate. This is not evidence of a mistaken physical-acceptance judgment. Neither the flag nor O PASS discharges inherited reporting obligations. The observed brownouts near 8766.18 and 19587.64 s occur after required active intervals; success on this waveform does not establish uninterrupted six-hour sensing [16, 17].

12.2 A frozen new waveform and complete denominators

A secondary closed comparison freezes six descendant design procedures before any final input. Fixed H0 is unchanged; G0 is a fixed procedure improver; G1 is one actually proposed improver. Three paired descendant lineages are conditional on that single G1 formation, not three independent G1 training runs. Thirty effective selection cells pass before freezing. Each final design episode then sees the complete current task, including its waveform. "Unexposed" describes formation and selection, not concealment of the waveform from the final designer [19].

Each arm has nine final cells: three lineage contexts on each of two reserved digital tasks and a new O waveform. Both digital tasks pass in all eighteen cells. The complete 27-cell totals are H0 7/1/1, G0 6/1/2 and G1 8/0/1 in PASS/submitted-FAIL/INCOMPLETE order. The O rows account for all differences (Tables 13 and 14). G1's 8/9 is descriptive conditional evidence, not a stable improvement claim or a reason to replace H0 [18, 21].

All five submitted new-O designs pass energy, match all eight events and receive 18 accepted packets. Three

Table 13: Complete 27-cell transfer denominator. Both reserved digital tasks pass in all 18 cells; O contributes the submitted failures and four incomplete episodes.

Procedure arm	PASS	Submitted FAIL	INCOMPLETE
H0	7	1	1
G0	6	1	2
G1	8	0	1

Table 14: All nine new-O final cells. Every submitted row matches eight events, receives 18 packets and passes energy. Events P/8 counts individually passing events. Dashes denote no submission.

Lineage/arm	Poll (s)	Retry (s)	Events P/8	Max error (s)	Service
L1-H0	1	1	6/8	1.103	FAIL
L1-G0	–	–	–	–	INCOMPLETE
L1-G1	0.75	0.25	8/8	0.728	PASS
L2-H0	0.75	5	8/8	0.728	PASS
L2-G0	–	–	–	–	INCOMPLETE
L2-G1	–	–	–	–	INCOMPLETE
L3-H0	–	–	–	–	INCOMPLETE
L3-G0	1	0.25	6/8	1.103	FAIL
L3-G1	0.75	1	8/8	0.728	PASS

designs sampled at 0.75 s pass complete service, with maximum encoded-time error 0.728 s. The two one-second submissions each pass six individual event checks and fail two stops with `WRONG_EVENT_TIME`. Their retry values differ, and their execution keys are distinct.

Table 15: The same two failed stop times occur in both one-second submissions. Error uses encoded event time, not receive time. Their distinct execution keys and different retry values remain preserved.

Confirmation (s)	Reception (s)	Encoded (s)	Error (s)
12077.625	12078.588	12078.629	1.004
17043.625	17044.588	17044.728	1.103

Both failed stop receptions occur less than one second after confirmation, but the encoded stamps are 1.004 and 1.103 s late (Table 15). Thus a correct match, prompt arrival, empty final queue and passing energy ledger cannot substitute for the encoded-time predicate. Detection and encoding must be checked together: for detection instant t_{det}, $|\hat{t} - t^*| \le |t_{\mathrm{det}} - t^*| + |\hat{t} - t_{\mathrm{det}}|$. The saved comparison does not isolate sampling phase from encoded clock behavior causally. Retry periods also vary, and one passing 0.75 s design belongs to H0. This is an observed design association, not a single-factor sampling ablation or an exclusive G1 mechanism.

Four O episodes have no submission: L1-G0, L2-G0, L2-G1 and L3-H0. Each last SDK assistant response reports stop reason `length`, 32,768 output tokens and 32,768 reasoning tokens, with thinking content and no final design or judgment. These are last-response counters, not the whole episode's token total or two additive token budgets. The outcomes remain INCOMPLETE; they are neither hardware failures nor energy FAILs. All requests, costs and nine O slots remain in the denominator, with no automatic retry [18].

The two submitted failures self-report PASS and therefore disagree with the nominal event predicate within the submitted scope. This differs from O0's broader UNKNOWN judgment. Missing submissions have no self-judgment and cannot be counted as zero-error successes.

12.3 New time-component transfer does not add an equal-history gain

A separate new finite component comparison uses three event-time tasks and two fresh contexts in each of four arms. It is ideally continuously powered; it tests event matching, timely reception and encoded-time accuracy

without the original energy, RF, active-keepalive or product obligations. All four arms pass all three predicates in all six formal attempts. H0, H0 with the same retained history, ordinary reflection and the proposed method use 28/26/43/29 requests, with conservative estimates of USD 0.1815672/0.6988500/1.1057187/0.7532616. Relative to the equal-history control, the proposed method improves neither success nor request count or estimated fee [27, 28].

The retained methods are actual model outputs formed from the same historical records and then frozen; formal outcomes do not revise them. All six ordinary and two proposed passing designs have at least one false frozen relation. The service verdict does not prove the method's semantic validity. Conversely, a conservative sufficient or applicability condition can be false on a feasible design without being unsound; these observations do not make the relations necessary conditions. Formation, validation and first transfer remain outside the twenty-four formal attempts, and the older four censored O episodes remain in their original denominator. This component result supports neither a new whole-device service pass nor stable recursive improvement.

13 Discussion and limitations

13.1 What changes when information is sufficient

The structural witnesses establish a precise failure of the current representation: identical present behavior can coexist with distinct responses to the same legal change. The enriched static controls show that the failure is not a universal static-versus-dynamic divide. Once the relevant law and state transition are supplied, the ideal tasks admit static prospective judgments. The matched comparison then tests actual finite-budget use, where the complete static arms already saturate service.

Live acquisition introduces more than information content. Source identity, source-local unit maps, domains, typed policy branches, proposal caps and final submission all become operational gates. The seven prospective E5 completions establish that this route can reach an actual consumer. Its five incomplete groups expose the cost of failing before execution. A compiler accepting a relation, an independent scalar agreement, a changed design and complete service are useful but different observations. None is automatically evidence of a necessary new fact or benefit over a sufficient static baseline.

The new eighteen-attempt comparison removes one ambiguity without removing this distinction. Cross-source preflight can admit a relation and three refinement attempts actually consume one. Nevertheless, every strong-control and refinement decision is already correct. The refinement path has equal acquisition and more requests than complete information, despite a slightly lower fee estimate. Actual usage of the interface is evidence that a software edge works, not evidence that it supplies information unavailable to a complete static model. The new saturated comparison cannot establish where such an advantage would emerge.

The O result further narrows what the consumer must check. A representation containing only matched counts and energy PASS would omit the distinction that separates the two one-second failures from complete event service. Adding receive delay alone would still miss the encoded-time failure. The relevant interface must expose the predicate's actual operands. This is a concrete consumer-specific information requirement, not a blanket request for a larger trace.

13.2 Limits on generalization and acceptance

The four main tasks are fixed public development inputs with three formal repetitions, not a random sample of task families. Pilot evidence and formal evidence remain separate. The same reported model alias is used, but pinned weight identity is UNKNOWN. Source and evaluator hashes support execution identity; they cannot recover unavailable weight identity or establish an invoice audit. The current study does not estimate statistical significance, format equivalence, general interface superiority or stable recursive improvement.

The new batch has only three fixed mechanisms and two fresh contexts per mechanism. These repeats are not six independent physical tasks or a sampled population. The source-access control is strong, but its initially supplied complete payload differs from the smaller catalogue-first contexts. Shared cache order and uncontrolled latency further limit a causal interpretation of token, fee or wall-time differences. We

therefore report each cost axis and preserve the two budget stops, rather than form a composite efficiency or autonomous-discovery score.

The ideal clamp and memory models share assumptions across their independent implementations. The transient source is prescribed, the passive latch is ideal, and the C-lite source has no measured RF conversion or load response. O adds proposed RMS costs and a modeled clock, not measured sensor, PMIC, NVM or PHY behavior. Range consistency, conservation checks, software correctness and physical qualification therefore remain distinct. A digital PASS does not qualify source spectra, startup transients, leakage, clock uncertainty, write persistence or radio reliability.

Original A/R dark-report and strict FIFO failures remain preserved, with product status `NOT_ACCEPTED`. O event success neither erases those obligations nor accepts a complete product [22]. Likewise, avoiding a reset is not retaining state through one, and matching an event is not encoding its time correctly. These scope distinctions prevent an attractive sub-result from answering a broader untested question.

13.3 The new finite product result remains negative

The accompanying A/R experiment uses a finite researcher-assembled component catalogue. The model compares a single storage module with a parallel pair, then selects one whole profile and executes its initial input plus two revisions: twelve original-service executions over four cases. The final candidate has independently consistent energy, receiver and activity traces, but all four complete service outcomes are FAIL. The light–dark case still passes dark new sampling; that local pass coexists with failures of early continuous sampling, first fresh reception and gap reporting. Product status remains `NOT_ACCEPTED`, and the component test-point projection remains `NOT_CALIBRATED` [26].

One cold-start exclusion has a sharper stated domain. For the selected single module, even its sourced 72 mF minimum capacitance and minimum 2.35 V ready threshold require

$$\tfrac{1}{2}(0.072)(2.35)^2 = 0.19881 \text{ J} > (90\ \mu\text{W})(1800 \text{ s}) = 0.162 \text{ J}. \tag{11}$$

Starting uncharged, this configuration cannot reach that threshold in time even with perfect conversion and no overhead. This exclusion does not depend on an assumed idle-current projection. It does not exclude every one- or two-module architecture: a series concept was neither selected nor executed, and startup isolation, balancing and qualified leakage remain open. The finite negative result supplies no global A/R impossibility claim. Likewise, the late FIFO recovery records cannot be attributed solely to energy without a separate causal control. No such additional experiment is introduced here.

13.4 Accounting and retained host interventions

The full matched campaign records 771 SETTLED requests, USD 5.4478605 in conservative rate estimates and 8914.281914 summed episode seconds. It includes the two method-proposal conditions outside the six displayed interface rows. E4 JSON is already the rich-static cell and is never added again. The secondary O/procedure campaign records 77 actual attempts for 73 logical cells, 367 SETTLED requests, USD 3.1767228 and 5951.639484 summed episode seconds. Its original O0 cost is a subset, not an additional campaign [11, 19, 21].

Four original selection host incidents are retained: three raise a missing historical-trace key and a fourth is censored by the hold. A registered continuation reconstructs the already sealed historical trace and links four named recovery attempts, preserving the original failures and valid earlier results. These infrastructure incidents are not negative G0/G1 scientific outcomes. Their 21 requests and USD 0.1318473 remain included in the 367-request total; they must not be added twice or silently removed [20].

All reported dollars are settled fixed-rate or conservative provider estimates, not invoice amounts. Historical UNKNOWN billing remains UNKNOWN in the coordinator's separate ledger. Codex assistance, historical acquisition and full end-to-end research cost are not priced by these episode sums. No hardware build or acceptance cost is inferred from them. Preparing this paper adds zero provider requests, scientific executions or native design mutations.

14 Saved-data reproducibility and implementation handoff

The preserved earlier package's `E_reproduce.py` uses only the Python standard library. It verifies that package's portable data manifest and reconstructs its eleven numerical tables, counts and timestamp differences. The new manuscript keeps those small frozen table sources and the full earlier argument, without copying the earlier raw-data tree or rerunning its experiments.

For the new increment, `extract_evidence.py` reads the canonical result index, frozen reports and already saved exact-input records. It verifies the new B report identity, extracts all eighteen rows, checks their aggregate counts against the report, and exports a deduplicated saved-trace view and source manifest. It imports no simulator, qualifier, consumer, evaluator, SDK or provider. `build_tables.py` and `build_figures.py` regenerate the new tables and vector figures from these small local views. Re-extraction requires the external canonical archive named by `-root`; the new TeX manuscript itself compiles independently from its included sources, tables, figures, style and logo. Standalone compilation is not a claim that all original raw experiments are duplicated in this small paper folder.

Original failures, post-hoc labels, runtime stops, prior UNKNOWN billing and product boundaries remain explicit. The official shared style and English vector mark are copied unchanged with readback hashes. The build receipt records actual repeated compilation, fonts, warnings and per-page visual inspection. All manuscript and figure preparation is read-only with respect to scientific inputs and contributes zero new model requests or scientific executions. No hardware acceptance, external submission, broader literature priority or venue-compliance review is claimed.

15 Conclusion

An interface supports a legal design change when it distinguishes the world differences relevant to that consumer's prospective verdict. Current success need not provide that information, while complete static laws can. In the matched digital comparison, strong static and executable inputs each pass all twelve formal groups; prospective observation completes seven and leaves five incomplete. The new three-mechanism comparison admits cross-source typed inputs and demonstrates real consumption, while all three arms make six correct decisions and select six passing designs. Refinement has equal acquisition and more requests than complete information; its slightly lower estimated fee is not comprehensive efficiency. Budget-stop labels and the actual submissions preceding them are both retained. An advantage over complete static information is therefore not established. The event-time study shows a further boundary: correct matching and energy can coexist with wrong encoded timestamps. The resulting evidence supports explicit information, admission and service predicates, with failures, costs and UNKNOWN retained. Physical qualification remains UNKNOWN, the original product remains NOT_ACCEPTED, and default H0 remains unchanged.

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